

```
tayyaba@tayyaba-VirtualBox: ~/xv6-labs-2026
tayyaba@tayyaba-VirtualBox:~$ cd ~/xv6-labs-2026
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ cd ~kernel/syscall.h
bash: cd: ~kernel/syscall.h: No such file or directory
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano kernel/syscall.h
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano kernel/syscall.c
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano kernel/sysproc.c
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano kernel/kalloc.c
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano kernel/defs.h
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano user/user.h
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano user/usys.pl
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano user/freememtest.c
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ nano Makefile
tayyaba@tayyaba-VirtualBox:~/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mv: cannot stat 'fs.img': No such file or directory
make: [Makefile:296: newfs.img] Error 1 (ignored)
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-poi
nter -ggdb -gdwarf-2 -ffile-prefix-map=/home/tayyaba/xv6-labs-2026=. -DSOL_UTIL -D
LAB_UTIL -march=rv64gc -std=gnu99 -MD -mcmmodel=medany -ffreestanding -fno-common -
nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-buili
n-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bz
ero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -f
no-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fpr
intf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o kernel/
kalloc.o kernel/kalloc.c
```

xv6 kernel is booting

hart 2 starting

hart 1 starting

init: starting sh

\$ freememtest

free memory: 133238784 bytes

\$ QEMU: Terminated

tayyaba@tayyaba-VirtualBox: ~/xv6-labs-2026\$